Rishab Gautam

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.03
Mount Litera Zee School Vidyaranyapura	Class XII (CBSE)	2024	92.80%
Nagarjuna Vidyaniketan	Class X (CBSE)	2022	92.00%

TECHNICAL ACHIEVEMENTS

Secured 3rd Place in **Vivriti Capital AI/ML Hackathon**, IIT Hyderabad

- **Built Intelli-Credit Copilot**, an end-to-end AI credit underwriting workbench that **converts raw financial documents** into a defensible **Credit Appraisal Memo** with lending recommendation, limit, and pricing.
- Engineered a **schema-guided extraction** pipeline combining a custom fast document runtime, geometry-aware parsing, candidate-page retrieval, **agentic fallback with LLM vision** for scanned pages, while preserving provenance back to source pages.

Secured 2nd place at **AVEVA National AI/ML Hackathon ,** IIT Hyderabad

- Architected a **multi-agent decision-support system** combining a Neo4j knowledge graph with **4 LLM-narrated agents** (risk briefing, similarity-based recommendations, carbon compliance, benchmark lifecycle management).

PROJECTS

Power Efficient CNN Accelerator on FPGA, *advised by Prof. Amit Acharyya, IIT Hyderabad*

- Designing an **FPGA-based CNN inference accelerator** targeting low-latency, power-efficient edge deployment.
- Established **GPU baselines** by profiling YOLOv11n and MobileNetV4 in FP32 and INT8 (latency, memory footprint, throughput) to guide accelerator design targets and quantization strategy.

Intelli-Credit Copilot: *3rd place,* ***Vivriti National AI/ML Hackathon*** Github

- Architected a full-stack loan underwriting platform **(FastAPI, async SQLAlchemy, Next.js 14)** covering onboarding, document upload, schema-guided extraction, Five Cs scoring, and automated **CAM report generation with export.**
- Designed a **3-tier document classifier** and a schema-driven table extractor with alias-based row/column matching, normalization, and **bbox-level provenance** so every extracted figure traces to an exact page location.

Rescaling vs. Routing: Testing the Attention Residuals Hypothesis at Scale Github

- Tested whether gains from the **Attention Residuals Research Paper** (Kimi/Moonshot AI, 2026) stem from learned depth-wise routing or from hidden-state magnitude rescaling, as hypothesized by Ziming Liu (2026).
- **Trained three 100M-parameter transformer variants** (baseline, rescaled residual, Block AttnRes) on FineWeb-Edu using the **Qwen3 architecture**; evaluated on WikiText-2, LAMBADA, and HellaSwag.
- Found parameter-free **rescaling matched or beat baseline** on HellaSwag/LAMBADA but underperformed on WikiText-2 PPL **(94.08 (baseline) vs. 90.3(rescaled))** (Block AttnRes: 88.89 PPL).

Built a GPT-2 LLM (124M params) from scratch in PyTorch Github

- Implemented and trained a **12-layer, 12-head, 768-dim decoder-only transformer** on the FineWeb-Edu dataset.
- **Benchmarked inference** on an RTX 3050 across FP32 and FP16 (half-precision) on GPU and FP32 on CPU, cutting VRAM from ~1.3GB to 682MB under FP16

Single-Cycle RISC-V Processor in Verilog Github

- Designed a **32-bit RISC-V processor** supporting RV32I ISA: ALU, register file, instruction decoder, and control unit.
- Built as a **single-cycle datapath** with unified control logic; verified instruction-level correctness by simulating hand-written RV32I assembly programs and inspecting register/memory state via GTKWave

POSITIONS OF RESPONSIBILITY

Tinkerers' Lab IITH Coordinator - Corporate Networking and Growth

- Coordinated Tinkerers' Lab's Flagship event's collaborative outreach.

TEDx IIT Hyderabad Coordinator - Public Relations and Networking Team

- Coordinated speaker outreach, sponsorship partnerships, and event logistics

SKILLS

Hardware: Icarus Verilog, EDAPlayground , Computer Architecture, RISC-V, Digital Design

Machine Learning: PyTorch, NumPy, Pandas, Scikit-learn, Hugging Face, LangChain

Languages: C++, C, Python, SQL, Verilog, MATLAB

RELEVANT COURSES

Software and ML: Data Structures and Applications, Introduction to Artificial Intelligence, Probability and Random Variables, Microprocessors and Computer Architecture, Introduction to Programming

Departmental core: Microprocessors and Computer Architecture, Linear Algebra, Matrix Theory, Digital IC design, Digital Systems, Numerical Methods for Device Modelling

EXTRACURRICULAR

Teaching Assistant: Engineering Electrodynamics by Prof. Aneesh S

- Conducted tutorial sessions and assisted students with problem-solving under Prof. Aneesh S.

Tennis-Inter IIT

- Represented IIT Hyderabad at Inter-IIT 2025 Sports Meet; former AITA Under-16 Open-tournament player.